Full Experiment Results for SoK: Certified Robustness for Deep Neural Networks

Abstract

In this document, we provide full experiment results for the paper "SoK: Certified Robustness for Deep Neural Networks", including detailed experimental setup and full results.

We evaluate and compare the robustness verification and training approaches by two sets of experiments: (1) for major deterministic verification approaches, we compare their certified robustness over a diverse set of trained models of different scales; (2) for major probabilistic verification approaches and their corresponding robust training approaches, we compare the best certified robustness they jointly achieve since the verification and training approaches are usually coupled. We do such separation mainly because in deterministic and probabilistic settings, the model inference methods are different and cannot be directly compared.

The full details for probabilistic set of experiments have been provided in the paper. Here, we present the *full experiment details* for the *deterministic verification approach benchmark*.

The open-sourced unified evaluation platform is available at *https://github.com/sokcertifiedrobustness/certified-robustness-benchmark*.

I. Experimental Setup

TABLE I: Deterministic verification approaches benchmarked in our evaluation.

Category	Approach	Implementation
MILP	`Bounded MILP` [20]	Reimplementation
Branch and Bound	`AI`^{2} [4]	From [16]
Linear Relaxation	`LP-Full` [21, 14]	From [1]
	`DeepPoly` [16]	From [16]
	`Fast-Lin` [21]	From [1]
	`CROWN` [24]	From [26]
	`CNN-Cert` [1]	From [1]
	`CROWN-IBP` [26]	From [26]
	`IBP` [5]	Reimplementation
	`WK` [22, 23]	From [23]
Hybrid	`k-ReLU` [15]	From [16]
	`RefineZono` [15]	From [16]
SDP	`SDPVerify` [13]	Reimplementation
	`LMIVerify` [3]	Reimplementation
Lipschitz	`Op-norm` [18]	From [25]
	`Fast-Lip` [21]	From [25]
	`RecurJac` [25]	From [25]

Evaluated Approaches. We present a thorough comparison of major *deterministic verification approaches* as listed in Table I. The list covers major branches of deterministic verification approaches.

Experiment Environment. For MNIST experiments, we run the evaluation on 24-core Intel Xeon E5-2650 CPU running at 2.20 GHz with single NVIDIA GeForce GTX 1080 Ti GPU. For CIFAR-10 experiments, we run the evaluation on 24-core Intel Xeon Platinum 8259CL CPU running at 2.50 GHz with single NVIDIA Tesla T4 GPU. The CIFAR-10 experiments use slightly faster running environment due to its larger scale.

Tool Implementation. We implement our tool mainly in `PyTorch` [12]. Some verification approaches use `keras` [7] and `Tensorflow` [19], for them, we implement a model translator from `PyTorch` directly to `keras` model, or to `onnx` [11] intermediate representation. The verification approaches reimplemented by ourselves use intrinsic `PyTorch` tensor computations, or `CVXPY` [2] + `Gurobi` [6] solver when there is an optimization problem to solve. We test our implementation to ensure the correctness and high-efficiency. For verification approaches with hyperparameters, we use their default hyperparameters as described in the papers or original implementations.

Dataset. We evaluate these verification approaches on image classification datasets MNIST [9] and CIFAR-10 [8].

Models. We include 7 different neural network structures from the literature and adapt them for each dataset. Among them, 3 (FCNNA - FCNNC) are fully-connected neural networks, and 4 (CNNA - CNND) are convolutional neural networks. They

TABLE II: Statistics of the models used in the evaluation of deterministic verification approaches. In the "Structure" column, the rightarrow $\rightarrow$ stands for the connection from previous to the next layer, "Flatten" stands for the flatten layer, "FC" stands for the fully connected layer, and "Conv" stands for the convolutional layer.

Dataset	Model	# Neurons	# Parameters	Structure	Source
MNIST	FCNNA	50	$16,330$	Flatten $\rightarrow$ 3 FCs	$2 \times [20]$ from [21, 25]
	FCNNB	310	$99,710$	Flatten $\rightarrow$ 4 FCs	$3 \times [100]$ enlarged from [21, 25]
	FCNNC	$7,178$	$7,111,690$	Flatten $\rightarrow$ 8 FCs	$7 \times [1024]$ from [21, 24]
	CNNA	$4,814$	$166,406$	2 Convs $\rightarrow$ Flatten $\rightarrow$ 2 FCs	Conv-Small in [22, 23]
	CNNB	$24,042$	$833,786$	4 Convs $\rightarrow$ Flatten $\rightarrow$ 2 FCs	Half sized Conv-Large in [23]
	CNNC	$48,074$	$1,974,762$	4 Convs $\rightarrow$ Flatten $\rightarrow$ 3 FCs	Conv-Large in [23]
	CNND	$176,138$	$13,314,634$	5 Convs $\rightarrow$ Flatten $\rightarrow$ 2 FCs	Double sized Conv-Large in [23]
CIFAR-10	FCNNA	50	$62,090$	Flatten $\rightarrow$ 3 FCs	$2 \times [20]$ from [21, 25]
	FCNNB	310	$328,510$	Flatten $\rightarrow$ 4 FCs	$3 \times [100]$ enlarged from [21, 25]
	FCNNC	$7,178$	$9,454,602$	Flatten $\rightarrow$ 8 FCs	$7 \times [1024]$ from [21, 24]
	CNNA	$6,254$	$214,918$	2 Convs $\rightarrow$ Flatten $\rightarrow$ 2 FCs	Conv-Small in [22, 23]
	CNNB	$31,242$	$1,079,834$	4 Convs $\rightarrow$ Flatten $\rightarrow$ 2 FCs	Half sized Conv-Large in [23]
	CNNC	$62,474$	$2,466,858$	4 Convs $\rightarrow$ Flatten $\rightarrow$ 3 FCs	Conv-Large in [23]
	CNND	$229,898$	$17,247,946$	5 Convs $\rightarrow$ Flatten $\rightarrow$ 2 FCs	Double sized Conv-Large in [23]

TABLE III: Clean accuracy and empirical robust accuracy for each set of weights. The clean accuracy is measured on the whole original test set. The empirical robust accuracy is measured on the inputs generated by PGD attacks from the whole original test set. For `reg` weights, due to low robustness, we do not report empirical robust accuracy (which is typically close to zero). On MNIST: for `adv1` and `cadv1`, the adversarial examples are bounded by radius $\epsilon = 0.1$ under $\mathcal{L}_\infty$ metric; for `adv3` and `cadv3`, bounded by radius $\epsilon = 0.3$. On CIFAR-10: for `adv2` and `cadv2`, bounded by radius $\epsilon = 2/255$; for `adv8` and `cadv8`, bounded by radius $\epsilon = 8/255$.

Dataset	Model	Weights	Clean Acc.	Empirical Rob. Acc.	Dataset	Model	Weights	Clean Acc.	Empirical Rob. Acc.
MNIST	FCNNA	`reg`	93.63%	/	CIFAR-10	FCNNA	`reg`	38.46%	/
		`adv1`	93.36%	78.39%			`adv2`	41.17%	39.49%
		`cadv1`	88.35%	80.99%			`cadv2`	38.97%	38.71%
		`adv3`	76.77%	31.67%			`adv8`	34.13%	31.36%
		`cadv3`	45.79%	41.35%			`cadv8`	30.59%	30.36%
	FCNNB	`reg`	96.12%	/		FCNNB	`reg`	41.76%	/
		`adv1`	97.12%	85.18%			`adv2`	44.31%	42.99%
		`cadv1`	95.23%	90.67%			`cadv2`	44.18%	43.83%
		`adv3`	89.96%	33.30%			`adv8`	37.68%	34.01%
		`cadv3`	76.21%	69.90%			`cadv8`	32.05%	31.40%
	FCNNC	`reg`	95.05%	/		FCNNC	`reg`	46.37%	/
		`adv1`	98.00%	87.40%			`adv2`	36.19%	35.87%
		`cadv1`	96.89%	93.43%			`cadv2`	46.34%	45.42%
		`adv3`	83.27%	33.31%			`adv8`	25.21%	25.87%
		`cadv3`	35.87%	34.08%			`cadv8`	30.30%	29.59%
	CNNA	`reg`	98.48%	/		CNNA	`reg`	59.45%	/
		`adv1`	99.01%	95.48%			`adv2`	60.40%	57.39%
		`cadv1`	98.52%	97.18%			`cadv2`	54.78%	54.17%
		`adv3`	98.20%	86.06%			`adv8`	49.42%	41.87%
		`cadv3`	96.26%	93.06%			`cadv8`	40.51%	39.93%
	CNNB	`reg`	98.85%	/		CNNB	`reg`	65.65%	/
		`adv1`	99.33%	96.09%			`adv2`	68.66%	64.01%
		`cadv1`	98.87%	98.12%			`cadv2`	58.81%	58.76%
		`adv3`	99.01%	92.84%			`adv8`	55.15%	45.35%
		`cadv3`	98.14%	95.88%			`cadv8`	40.05%	39.36%
	CNNC	`reg`	98.85%	/		CNNC	`reg`	58.60%	/
		`adv1`	99.24%	96.35%			`adv2`	65.59%	62.05%
		`cadv1`	98.83%	98.41%			`cadv2`	59.46%	59.25%
		`adv3`	99.16%	93.47%			`adv8`	54.01%	45.91%
		`cadv3`	98.11%	96.10%			`cadv8`	40.11%	39.50%
	CNND	`reg`	99.20%	/		CNND	`reg`	83.53%	/
		`adv1`	99.37%	97.64%			`adv2`	83.65%	76.41%
		`cadv1`	99.13%	98.50%			`cadv2`	60.74%	59.88%
		`adv3`	99.38%	95.09%			`adv8`	72.25%	53.86%
		`cadv3`	98.58%	96.72%			`cadv8`	40.61%	39.94%

are all feed-forward ReLU neural networks. The number of neurons ranges from 50 (FCNNA) to about $200,000$ (CNND). For each neural network structure, we train 5 sets of weights:

- `clean` (on both MNIST and CIFAR-10): regular training;
- `adv1` (on MNIST)/`adv2` (on CIFAR-10): PGD adversarial training with $\epsilon = 0.1$ (on MNIST) or $\epsilon = 2/255$ (on CIFAR-10);
- `adv3` (on MNIST)/`adv8` (on CIFAR-10): PGD adversarial training with $\epsilon = 0.3$ (on MNIST) or $\epsilon = 8/255$ (on CIFAR-10);
- `cadv1` (on MNIST)/`cadv2` (on CIFAR-10): `CROWN-IBP` training with $\epsilon = 0.1$ (on MNIST) or $\epsilon = 2/255$ (on CIFAR-10);
- `cadv3` (on MNIST)/`cadv8` (on CIFAR-10): `CROWN-IBP` training with $\epsilon = 0.3$ (on MNIST) or $\epsilon = 8/255$ (on CIFAR-10).

Here, ϵ is the $\mathcal{L}_\infty$ radius of the adversarial examples. The PGD adversarial training [10] is a strong empirical defense. We use 40-step PGD in PGD adversarial training. The `CROWN-IBP` [26] is a strong robust training approach which achieves state-of-the-art certified robustness under $(\mathcal{L}_\infty, 8/255)$ adversary. We choose these training configurations to reflect 3 common types of models on which verification approaches are used: normal models; empirical defense models; and robustly trained models. All models are trained to reach their expected accuracy or robustness. The model structures and statistics of number of neurons is shown in Table II. The model accuracy on original test set, and empirical robust accuracy under the PGD attack, is shown in Table III.

Evaluation protocol. We present the performance of verification approaches by their *robust accuracy* and *average certified robustness radius* with respect to $\mathcal{L}_\infty$ radius ϵ.

Robust Accuracy: When measuring robust accuracy, the chosen radius ϵ for verification corresponds to the radius trained for defense. The robust accuracy is defined as $\text{robacc} := \frac{\text{\# samples verified to be robust}}{\text{\# number of all samples}}$. On each dataset, we uniformly sample 100 test set samples as the fixed set for evaluation. We further limit the running time to $60\,\text{s}$ per instance and count timeout instances as "non-robust".

We also report the robust accuracy under empirical attack (PGD attack), which gives an upper bound of robust accuracy, so we can estimate the gap between certified robust accuracy and accuracy of existing attack. We remark that due to hyperparameter tuning, limited running time ($60\,\text{s}$), and different neural network models used in evaluation, the robust accuracy of some approaches differs much with the reported numbers, such as `k-ReLU` [15], `RefineZono` [17], `SDPVerify` [13], and `LMIVerify` [3].

Average Certified Robustness Radius: We also evaluate the verification approaches by measuring their average certified robustness radius. The *average certified robustness radius* ($\bar{r}$) stands for the average $\mathcal{L}_\infty$ radius the verification approach can verify on the given subset of test set samples. We use the same uniformly sampled sets as in robust accuracy evaluation. To determine the best certified radius of each verification approach, we conduct a binary search process due to the monotonicity. Specifically, we do binary search on interval $[0,\, 0.5]$ because the largest possible radius is 0.5 for $[0,\, 1]$ bounded input. If current radius mid is verified to be robust, we update current best by mid and let $l \leftarrow mid$, if current radius mid cannot be verified, we let $r \leftarrow mid$, until we reach the precision 10^{-2} on MNIST or 10^{-3} on CIFAR-10, or time is up. For the evaluation of average certified robustness radius, since it involves multiple evaluations because of binary search, We limit the running time to $120\,\text{s}$ per input and record the highest certified radius it has been verified before the time is used up. The average certified robustness radius is evaluated on the same subset of test set samples as used for robust accuracy evaluation.

We also report the smallest radius of adversarial samples found by empirical attack (PGD attack), which gives an upper bound of certified robustness radius for us to estimate the gap.

Throughout the evaluation, we use 100-step PGD attack with step size equals to $\epsilon/50$ where ϵ is the constrained $\mathcal{L}_\infty$ attack radius.

II. Full Results

We present the full results as below. Besides robust accuracy and average certified robustness radius, we also report the running time per instance for each approach on both tasks respectively. Note that for robust accuracy evaluation, the verification is ran once per instance with time limit $60\,\text{s}$; while for average certified robustness radius evaluation, the verification will be ran multiple times due to the binary search, with overall time limit $120\,\text{s}$.

We summarize the references of corresponding result tables as below.

Dataset	Evaluation Type	Results Table	Running Time Table
MNIST	Robust Accuracy (robacc)	Table IV	Table V
	Average certified robustness radius ($\bar{r}$)	Table VI	Table VII
CIFAR-10	Robust Accuracy (robacc)	Table VIII[1]	Table IX
	Average certified robustness radius ($\bar{r}$)	Table X	Table XI

Remarks.

Generally, the tendency on MNIST is the same as that on CIFAR-10.

From these additional evaluations of average certified robustness radius, we can find that the average radius has better precision than robust accuracy. For example, on small models such as FNNA and FNNB, if measured by robust accuracy, the complete verification approaches and some linear relaxation-based approaches have almost the same precision (see table VIII,

[1]The same table as appeared in the main paper.

TABLE IV: *Robust accuracy* on MNIST models certified by different *deterministic verification approaches*. The verification is under $\mathcal{L}_\infty$ adversary with radius ϵ (specified in table). We include accuracy under PGD attack to serve as the upper bound of robust accuracy, and the clean accuracy. The **bolded** numbers mark the highest ones among verification approaches.

$\epsilon = 0.02$	FCNNA	FCNNB	FCNNC	CNNA	CNNB	CNNC	CNND
	clean	clean	clean	clean	clean	clean	clean
Bounded MILP	**80**%	84%	0%	44%	0%	0%	0%
AI^2	**80**%	**86**%	0%	**89**%	**90**%	0%	0%
LP-Full	70%	68%	0%	0%	0%	0%	0%
DeepPoly	79%	85%	0%	**89**%	**90**%	0%	0%
Fast-Lin	68%	56%	0%	72%	28%	**5**%	0%
CROWN	79%	85%	0%	**89**%	**90**%	0%	0%
CNN-Cert	70%	67%	0%	73%	44%	0%	0%
CROWN-IBP	77%	0%	0%	13%	0%	0%	0%
IBP	3%	0%	0%	0%	0%	0%	0%
WK	68%	56%	0%	72%	28%	**5**%	0%
k-ReLU	79%	85%	0%	**89**%	0%	0%	0%
RefineZono	79%	0%	0%	0%	0%	0%	0%
SDPVerify	0%	0%	0%	0%	0%	0%	0%
LMIVerify	0%	0%	0%	0%	0%	0%	0%
Op-norm	0%	0%	0%	0%	0%	0%	0%
FastLip	60%	4%	0%	6%	0%	0%	0%
RecurJac	70%	52%	0%	33%	0%	0%	0%
PGD Upper Bound	80%	86%	85%	92%	95%	93%	97%
Clean Acc.	89%	92%	95%	99%	100%	100%	100%

$\epsilon = 0.1$	FCNNA		FCNNB		FCNNC		CNNA		CNNB		CNNC		CNND	
	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1
Bounded MILP	**70**%	**68**%	0%	**85**%	0%	67%	0%	**95**%	0%	0%	0%	0%	0%	0%
AI^2	**70**%	**68**%	1%	**85**%	0%	52%	6%	**95**%	1%	93%	0%	0%	0%	0%
LP-Full	8%	46%	0%	57%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
DeepPoly	47%	64%	**4**%	72%	0%	52%	**32**%	93%	**2**%	93%	0%	0%	0%	0%
Fast-Lin	5%	35%	0%	36%	0%	0%	0%	90%	0%	89%	0%	86%	0%	0%
CROWN	47%	64%	**4**%	72%	0%	52%	**32**%	93%	**2**%	93%	0%	0%	0%	0%
CNN-Cert	5%	36%	0%	44%	0%	0%	0%	92%	0%	93%	0%	0%	0%	0%
CROWN-IBP	2%	67%	0%	76%	0%	73%	0%	93%	0%	**94**%	0%	88%	0%	**94**%
IBP	0%	**68**%	0%	**85**%	0%	**80**%	0%	**95**%	0%	91%	0%	**89**%	0%	90%
WK	5%	35%	0%	36%	0%	0%	0%	90%	0%	89%	0%	86%	0%	1%
k-ReLU	51%	**68**%	**4**%	**85**%	0%	0%	3%	**95**%	0%	0%	0%	0%	0%	0%
RefineZono	49%	63%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
SDPVerify	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
LMIVerify	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Op-norm	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
FastLip	0%	30%	0%	31%	0%	2%	0%	86%	0%	0%	0%	0%	0%	0%
RecurJac	2%	30%	0%	31%	0%	0%	0%	85%	0%	0%	0%	0%	0%	0%
PGD Upper Bound	71%	69%	80%	85%	82%	89%	94%	95%	95%	97%	95%	96%	97%	99%
Clean Acc.	92%	88%	97%	97%	98%	96%	100%	99%	100%	99%	100%	100%	100%	100%

$\epsilon = 0.3$	FCNNA		FCNNB		FCNNC		CNNA		CNNB		CNNC		CNND	
	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3
Bounded MILP	6%	**25**%	0%	**54**%	0%	7%	0%	**88**%	0%	84%	0%	0%	0%	0%
AI^2	6%	**25**%	0%	**54**%	0%	16%	0%	73%	0%	0%	0%	0%	0%	0%
LP-Full	0%	15%	0%	8%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
DeepPoly	0%	23%	0%	27%	0%	16%	0%	40%	0%	7%	0%	0%	0%	0%
Fast-Lin	0%	4%	0%	1%	0%	0%	0%	6%	0%	0%	0%	0%	0%	0%
CROWN	0%	23%	0%	27%	0%	16%	0%	40%	0%	7%	0%	0%	0%	0%
CNN-Cert	0%	13%	0%	5%	0%	0%	0%	39%	0%	1%	0%	0%	0%	0%
CROWN-IBP	0%	23%	0%	45%	0%	20%	0%	76%	0%	71%	0%	75%	0%	65%
IBP	0%	**25**%	0%	**54**%	0%	**22**%	0%	**88**%	0%	**85**%	0%	**87**%	0%	**89**%
WK	0%	4%	0%	1%	0%	0%	0%	6%	0%	0%	0%	0%	0%	0%
k-ReLU	0%	**25**%	0%	**54**%	0%	0%	0%	57%	0%	0%	0%	0%	0%	0%
RefineZono	0%	21%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
SDPVerify	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
LMIVerify	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Op-norm	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
FastLip	0%	1%	0%	2%	0%	0%	0%	2%	0%	0%	0%	0%	0%	0%
RecurJac	0%	1%	0%	2%	0%	0%	0%	2%	0%	0%	0%	0%	0%	0%
PGD Upper Bound	19%	26%	26%	64%	26%	26%	81%	90%	91%	94%	90%	93%	92%	94%
Clean Acc.	75%	37%	88%	76%	78%	28%	97%	95%	100%	98%	99%	98%	100%	98%

TABLE V: *Average running time for single-instance robustness verification* in seconds per correctly-predicted instance on MNIST models of different verification approaches. The verification is under $\mathcal{L}_\infty$ adversary with ϵ (specified in table) radius. We stop the execution when time exceeds 60 s per instance. We include running time of PGD attack and normal inference as the reference.

	FCNNA	FCNNB	FCNNC	CNNA	CNNB	CNNC	CNND
	clean	clean	clean	clean	clean	clean	clean
Bounded MILP	0.37	12.67	60.00	54.28	60.00	60.00	60.00
AI^2	0.09	1.09	60.00	7.92	48.89	60.00	60.00
LP-Full	0.41	15.22	60.00	60.00	60.00	60.00	60.00
DeepPoly	0.03	0.25	60.00	2.45	49.04	60.00	60.00
Fast-Lin	0.03	0.03	1.71	0.35	4.70	15.95	60.00
CROWN	0.01	0.01	0.03	0.02	0.13	60.00	60.00
CNN-Cert	0.12	0.43	60.00	1.13	28.40	59.67	60.00
CROWN-IBP	0.01	0.01	0.01	0.01	0.01	0.01	0.01
IBP	0.00	0.00	0.01	0.00	0.01	0.01	0.01
WK	0.01	0.01	0.04	0.01	0.03	0.08	60.00
k-ReLU	12.49	26.05	60.00	30.49	60.00	60.00	60.00
RefineZono	3.40	60.00	60.00	60.00	60.00	60.00	60.00
SDPVerify	60.00	60.00	60.00	60.00	60.00	60.00	60.00
LMIVerify	60.00	60.00	60.00	60.00	60.00	60.00	60.00
Op-norm	0.00	0.00	0.00	0.00	0.01	0.01	60.00
FastLip	0.20	0.20	1.68	5.81	25.47	54.26	60.00
RecurJac	0.81	2.25	43.29	50.35	60.00	60.00	60.00
PGD Attack	0.26	0.30	0.42	0.33	0.43	0.46	0.51
Normal Inference	0.00	0.00	0.00	0.00	0.00	0.00	0.00

	FCNNA		FCNNB		FCNNC		CNNA		CNNB		CNNC		CNND	
	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1
Bounded MILP	0.73	0.28	60.00	1.32	60.00	53.35	60.00	6.86	60.00	60.00	60.00	60.00	60.00	60.00
AI^2	0.52	0.23	59.38	1.12	60.00	31.50	56.49	2.61	59.80	25.03	60.00	60.00	60.00	60.00
LP-Full	0.43	0.35	23.59	9.57	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
DeepPoly	0.03	0.02	0.24	0.15	60.00	7.97	1.61	1.34	41.59	23.53	60.00	60.00	60.00	60.00
Fast-Lin	0.01	0.01	0.03	0.02	2.06	1.71	0.39	0.41	5.63	4.80	17.31	19.13	60.00	60.00
CROWN	0.01	0.01	0.01	0.01	0.03	0.03	0.02	0.02	0.13	0.13	60.00	60.00	60.00	60.00
CNN-Cert	0.02	0.03	0.05	0.31	60.00	60.00	0.21	1.35	4.18	36.00	60.00	60.00	60.00	60.00
CROWN-IBP	0.00	0.00	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
IBP	0.00	0.00	0.00	0.01	0.01	0.01	0.00	0.01	0.01	0.01	0.01	0.01	0.01	0.01
WK	0.01	0.01	0.01	0.01	0.04	0.04	0.01	0.01	0.07	0.03	0.21	0.09	60.00	59.41
k-ReLU	12.87	11.92	34.56	38.43	60.00	60.00	59.35	34.48	60.00	60.00	60.00	60.00	60.00	60.00
RefineZono	7.27	14.51	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
SDPVerify	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
LMIVerify	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
Op-norm	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	60.00	60.00
FastLip	0.02	0.03	0.05	0.12	1.17	6.04	2.52	5.51	20.05	60.00	46.69	60.00	60.00	60.00
RecurJac	0.36	0.35	0.88	1.77	24.96	57.75	28.61	49.91	60.00	60.00	60.00	60.00	60.00	60.00
PGD Attack	0.26	0.30	0.30	0.33	0.42	0.48	0.33	0.37	0.44	0.48	0.47	0.50	0.52	0.54
Normal Inference	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

	FCNNA		FCNNB		FCNNC		CNNA		CNNB		CNNC		CNND	
	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3
Bounded MILP	0.63	0.30	60.00	1.39	60.00	52.52	60.00	5.61	60.00	50.35	60.00	60.00	60.00	60.00
AI^2	0.72	0.29	60.00	1.72	60.00	23.66	60.00	27.46	60.00	60.00	60.00	60.00	60.00	60.00
LP-Full	0.49	0.39	22.97	12.75	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
DeepPoly	0.02	0.02	0.29	0.15	60.00	5.14	1.75	1.28	40.27	27.57	60.00	60.00	60.00	60.00
Fast-Lin	0.01	0.01	0.03	0.03	2.14	1.96	0.34	0.44	5.66	5.28	18.63	20.04	60.00	60.00
CROWN	0.01	0.01	0.01	0.01	0.03	0.03	0.02	0.02	0.13	0.13	60.00	60.00	60.00	60.00
CNN-Cert	0.01	0.02	0.06	0.12	60.00	60.00	0.17	0.89	4.20	5.91	60.00	60.00	60.00	60.00
CROWN-IBP	0.01	0.01	0.01	0.01	0.02	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.02	0.01
IBP	0.00	0.00	0.00	0.01	0.01	0.01	0.00	0.01	0.01	0.01	0.01	0.01	0.01	0.01
WK	0.01	0.01	0.01	0.01	0.04	0.04	0.01	0.01	0.07	0.06	0.38	0.19	60.00	60.00
k-ReLU	12.37	12.78	45.15	44.37	60.00	60.00	60.00	48.59	60.00	60.00	60.00	60.00	60.00	60.00
RefineZono	12.08	18.52	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
SDPVerify	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
LMIVerify	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
Op-norm	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.01	60.00	60.00
FastLip	0.01	0.03	0.02	0.07	1.87	1.91	2.04	3.11	8.27	50.14	46.81	60.00	60.00	60.00
RecurJac	0.14	0.31	0.36	1.28	10.81	29.23	12.47	38.94	60.00	60.00	60.00	60.00	60.00	60.00
PGD Attack	0.26	0.32	0.30	0.34	0.43	0.49	0.33	0.38	0.45	0.48	0.47	0.51	0.52	0.54
Normal Inference	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

TABLE VI: *Average certified robustness radius* on MNIST models certified by different *deterministic verification approaches*. The verification is under $\mathcal{L}_\infty$ adversary. We include radius of adversarial example found by PGD attack to serve as the upper bound. The **bolded** numbers mark the highest ones among verification approaches.

	FCNNA	FCNNB	FCNNC	CNNA	CNNB	CNNC	CNND
	clean	clean	clean	clean	clean	clean	clean
Bounded MILP	**0.054**	0.000	0.000	0.000	0.000	0.000	0.000
AI^2	**0.054**	0.000	0.000	0.000	0.000	0.000	0.000
LP-Full	0.027	0.019	0.000	0.000	0.000	0.000	0.000
DeepPoly	0.045	**0.035**	0.000	**0.039**	0.000	0.000	0.000
Fast-Lin	0.026	0.018	0.000	0.021	0.014	**0.009**	0.000
CROWN	0.045	**0.035**	**0.005**	**0.039**	**0.030**	0.000	0.000
CNN-Cert	0.027	0.020	0.000	0.022	0.015	0.000	0.000
CROWN-IBP	0.033	0.006	0.000	0.011	0.000	0.000	0.000
IBP	0.006	0.000	0.000	0.000	0.000	0.000	0.000
WK	0.026	0.018	0.000	0.021	0.014	**0.009**	0.000
k-ReLU	0.046	0.001	0.000	0.000	0.000	0.000	0.000
RefineZono	0.045	0.000	0.000	0.000	0.000	0.000	0.000
SDPVerify	0.000	0.000	0.000	0.000	0.000	0.000	0.000
LMIVerify	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Op-norm	0.003	0.001	0.000	0.000	0.000	0.000	0.000
FastLip	0.020	0.008	0.000	0.009	0.000	0.000	0.000
RecurJac	0.030	0.021	0.000	0.009	0.000	0.000	0.000
PGD Upper Bound	0.057	0.060	0.046	0.073	0.087	0.090	0.103

	FCNNA		FCNNB		FCNNC		CNNA		CNNB		CNNC		CNND	
	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1
Bounded MILP	**0.135**	**0.131**	0.000	**0.150**	0.000	0.000	0.000	0.122	0.000	0.000	0.000	0.000	0.000	0.000
AI^2	**0.135**	**0.131**	0.000	**0.150**	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
LP-Full	0.063	0.097	0.040	0.102	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
DeepPoly	0.095	0.120	**0.064**	0.118	0.000	0.096	**0.086**	0.152	0.008	0.160	0.000	0.000	0.000	0.000
Fast-Lin	0.060	0.083	0.036	0.080	0.007	0.056	0.061	0.142	0.051	0.149	**0.041**	0.138	0.000	0.000
CROWN	0.095	0.120	**0.064**	0.118	**0.019**	0.096	**0.086**	0.152	**0.075**	0.168	0.000	0.000	0.000	0.000
CNN-Cert	0.062	0.084	0.039	0.085	0.000	0.000	0.066	0.150	0.056	0.161	0.000	0.000	0.000	0.000
CROWN-IBP	0.055	0.123	0.009	0.135	0.000	0.133	0.030	**0.154**	0.001	**0.176**	0.000	**0.179**	0.000	**0.174**
IBP	0.011	0.128	0.000	0.142	0.000	**0.143**	0.005	**0.154**	0.000	0.174	0.000	0.175	0.000	0.168
WK	0.060	0.083	0.036	0.080	0.007	0.056	0.061	0.142	0.051	0.149	**0.041**	0.139	0.000	0.000
k-ReLU	0.100	0.129	0.014	0.097	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
RefineZono	0.097	0.120	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
SDPVerify	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
LMIVerify	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Op-norm	0.002	0.002	0.001	0.002	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
FastLip	0.041	0.078	0.016	0.074	0.000	0.052	0.032	0.131	0.000	0.000	0.000	0.000	0.000	0.000
RecurJac	0.060	0.081	0.035	0.078	0.000	0.000	0.032	0.107	0.000	0.000	0.000	0.000	0.000	0.000
PGD Upper Bound	0.145	0.157	0.155	0.190	0.151	0.213	0.163	0.194	0.168	0.233	0.169	0.278	0.181	0.320

	FCNNA		FCNNB		FCNNC		CNNA		CNNB		CNNC		CNND	
	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3
Bounded MILP	**0.157**	**0.256**	0.000	**0.278**	0.000	0.230	0.000	0.255	0.000	0.233	0.000	0.018	0.000	0.000
AI^2	**0.157**	**0.256**	0.000	**0.278**	0.000	0.232	0.000	0.240	0.000	0.000	0.000	0.000	0.000	0.000
LP-Full	0.079	0.229	0.049	0.217	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
DeepPoly	0.119	0.239	**0.076**	0.224	0.000	0.228	**0.174**	0.254	0.105	0.188	0.000	0.000	0.000	0.000
Fast-Lin	0.074	0.202	0.043	0.151	0.008	0.033	0.117	0.221	0.106	0.174	0.083	0.143	0.000	0.000
CROWN	0.119	0.239	**0.076**	0.224	**0.021**	0.228	**0.174**	0.254	**0.141**	0.204	0.000	0.000	0.000	0.000
CNN-Cert	0.078	0.210	0.048	0.189	0.000	0.000	0.130	0.253	0.120	0.179	0.000	0.000	0.000	0.000
CROWN-IBP	0.069	0.239	0.005	0.256	0.000	0.288	0.074	0.283	0.000	0.272	0.000	0.277	0.000	0.272
IBP	0.008	**0.256**	0.000	0.275	0.000	**0.295**	0.018	**0.294**	0.000	**0.296**	0.000	**0.292**	0.000	**0.299**
WK	0.074	0.202	0.043	0.151	0.008	0.033	0.117	0.221	0.106	0.174	**0.085**	0.143	0.000	0.000
k-ReLU	0.125	**0.256**	0.002	0.209	0.000	0.000	0.000	0.227	0.000	0.000	0.000	0.000	0.000	0.000
RefineZono	0.120	0.237	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
SDPVerify	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
LMIVerify	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Op-norm	0.001	0.000	0.000	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
FastLip	0.053	0.189	0.015	0.153	0.000	0.043	0.067	0.199	0.000	0.000	0.000	0.000	0.000	0.000
RecurJac	0.072	0.193	0.039	0.157	0.000	0.000	0.058	0.146	0.000	0.000	0.000	0.000	0.000	0.000
PGD Upper Bound	0.199	0.309	0.216	0.331	0.231	0.427	0.308	0.302	0.313	0.318	0.328	0.316	0.366	0.325

TABLE VII: *Average running time for robustness radius computation* in seconds per correctly-predicted instance on MNIST models of different verification approaches. The verification is on $\mathcal{L}_\infty$ ball. We stop the execution when time exceeds 120 s per instance. We include running time of PGD attack as the reference.

	FCNNA	FCNNB	FCNNC	CNNA	CNNB	CNNC	CNND
	clean	clean	clean	clean	clean	clean	clean
Bounded MILP	3.14	120.00	120.00	120.00	120.00	120.00	120.00
AI^2	3.95	120.00	120.00	120.00	120.00	120.00	120.00
LP-Full	2.63	113.84	120.00	120.00	120.00	120.00	120.00
DeepPoly	0.17	1.50	120.00	13.42	120.00	120.00	120.00
Fast-Lin	0.04	0.17	12.30	2.04	32.25	103.94	120.00
CROWN	0.03	0.05	0.17	0.09	0.80	120.00	120.00
CNN-Cert	0.11	1.19	120.00	3.23	75.62	120.00	120.00
CROWN-IBP	0.03	0.03	0.07	0.04	0.06	0.08	0.08
IBP	0.02	0.02	0.03	0.02	0.03	0.04	0.05
WK	0.04	0.06	0.22	0.08	0.32	1.06	120.00
k-ReLU	70.72	120.00	120.00	120.00	120.00	120.00	120.00
RefineZono	32.98	120.00	120.00	120.00	120.00	120.00	120.00
SDPVerify	120.00	120.00	120.00	120.00	120.00	120.00	120.00
LMIVerify	120.00	120.00	120.00	120.00	120.00	120.00	120.00
Op-norm	0.00	0.00	0.00	0.00	0.00	0.01	120.00
FastLip	0.15	0.57	9.95	19.23	111.50	120.00	120.00
RecurJac	6.81	29.75	120.00	114.36	120.00	120.00	120.00
PGD Attack	1.61	1.81	2.59	2.03	2.68	2.85	3.13

	FCNNA		FCNNB		FCNNC		CNNA		CNNB		CNNC		CNND	
	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1	adv1	cadv1
Bounded MILP	4.97	2.39	120.00	14.06	120.00	120.00	120.00	118.88	120.00	120.00	120.00	120.00	120.00	120.00
AI^2	5.28	6.44	120.00	39.72	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
LP-Full	2.77	2.33	107.16	73.58	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
DeepPoly	0.16	0.12	1.50	0.99	120.00	65.62	9.55	8.65	120.00	120.00	120.00	120.00	120.00	120.00
Fast-Lin	0.03	0.04	0.16	0.17	11.18	9.15	2.11	2.62	32.73	28.51	96.23	119.36	120.00	120.00
CROWN	0.03	0.03	0.05	0.05	0.17	0.18	0.09	0.09	0.80	0.80	120.00	120.00	120.00	120.00
CNN-Cert	0.16	0.16	1.48	1.70	120.00	120.00	4.09	4.69	94.97	115.68	120.00	120.00	120.00	120.00
CROWN-IBP	0.03	0.03	0.03	0.03	0.10	0.07	0.04	0.04	0.06	0.07	0.08	0.08	0.08	0.08
IBP	0.02	0.02	0.02	0.03	0.03	0.04	0.02	0.03	0.03	0.04	0.04	0.05	0.05	0.06
WK	0.04	0.04	0.06	0.06	0.21	0.20	0.08	0.08	0.33	0.23	1.03	0.77	120.00	120.00
k-ReLU	65.10	69.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
RefineZono	56.38	100.87	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
SDPVerify	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
LMIVerify	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
Op-norm	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	120.00	120.00
FastLip	0.16	0.16	0.67	0.69	8.59	46.72	20.61	25.84	120.00	120.00	120.00	120.00	120.00	120.00
RecurJac	5.87	5.62	29.77	28.10	120.00	120.00	119.21	120.00	120.00	120.00	120.00	120.00	120.00	120.00
PGD Attack	1.70	2.04	1.91	2.25	2.67	2.96	2.11	2.33	2.74	2.97	2.85	3.21	3.16	3.40

	FCNNA		FCNNB		FCNNC		CNNA		CNNB		CNNC		CNND	
	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3	adv3	cadv3
Bounded MILP	4.75	2.53	120.00	7.77	120.00	120.00	120.00	116.93	120.00	120.00	120.00	120.00	120.00	120.00
AI^2	4.96	3.74	120.00	12.73	120.00	117.14	120.00	118.62	120.00	120.00	120.00	120.00	120.00	120.00
LP-Full	2.51	2.39	104.91	63.49	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
DeepPoly	0.16	0.13	1.76	0.94	120.00	32.34	10.85	8.12	120.00	120.00	120.00	120.00	120.00	120.00
Fast-Lin	0.03	0.04	0.17	0.15	11.21	8.92	1.78	2.40	29.16	28.74	115.24	106.29	120.00	120.00
CROWN	0.03	0.03	0.05	0.05	0.17	0.18	0.09	0.09	0.80	0.80	120.00	120.00	120.00	120.00
CNN-Cert	0.16	0.16	1.41	1.95	120.00	120.00	4.71	5.14	112.83	117.19	120.00	120.00	120.00	120.00
CROWN-IBP	0.03	0.03	0.03	0.03	0.07	0.08	0.04	0.04	0.10	0.07	0.08	0.08	0.08	0.08
IBP	0.02	0.02	0.02	0.03	0.03	0.04	0.02	0.03	0.03	0.04	0.04	0.05	0.05	0.06
WK	0.04	0.04	0.06	0.06	0.21	0.18	0.07	0.08	0.18	0.21	0.57	0.78	120.00	120.00
k-ReLU	64.68	63.76	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
RefineZono	79.13	21.92	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
SDPVerify	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
LMIVerify	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
Op-norm	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.01	0.01	120.00	120.00
FastLip	0.16	0.19	0.62	0.74	9.15	38.31	31.29	25.23	120.00	120.00	120.00	120.00	120.00	120.00
RecurJac	5.37	5.92	29.40	30.22	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
PGD Attack	1.81	2.38	2.00	2.23	2.79	3.08	2.10	2.24	2.71	2.82	2.92	3.10	3.25	3.32

TABLE VIII: *Robust accuracy* on CIFAR-10 models certified by different *deterministic verification approaches*. The verification is under $\mathcal{L}_\infty$ adversary with radius ϵ (specified in table). We include accuracy under PGD attack to serve as the upper bound of robust accuracy, and the clean accuracy. The **bolded** numbers mark the highest ones among verification approaches.

$\epsilon = 0.5/255$	FCNNA	FCNNB	FCNNC	CNNA	CNNB	CNNC	CNND
	clean	clean	clean	clean	clean	clean	clean
Bounded MILP	**40%**	**43%**	0%	23%	0%	0%	0%
AI^2	**40%**	**43%**	0%	**51%**	9%	0%	0%
LP-Full	**40%**	**43%**	0%	0%	0%	0%	0%
DeepPoly	**40%**	**43%**	0%	**51%**	**52%**	0%	0%
Fast-Lin	**40%**	42%	49%	**51%**	49%	**45%**	0%
CROWN	**40%**	**43%**	**51%**	**51%**	**52%**	0%	0%
CNN-Cert	**40%**	**43%**	0%	**51%**	**52%**	0%	0%
CROWN-IBP	**40%**	24%	0%	13%	0%	0%	0%
IBP	29%	0%	0%	0%	0%	0%	0%
WK	**40%**	42%	49%	**51%**	49%	**45%**	**3%**
k-ReLU	**40%**	**43%**	0%	48%	0%	0%	0%
RefineZono	0%	0%	0%	0%	0%	0%	0%
SDPVerify	0%	0%	0%	0%	0%	0%	0%
LMIVerify	0%	0%	0%	0%	0%	0%	0%
Op-norm	0%	0%	0%	0%	0%	0%	0%
FastLip	**40%**	40%	0%	36%	0%	0%	0%
RecurJac	**40%**	42%	0%	0%	0%	0%	0%
PGD Upper Bound	40%	43%	53%	51%	61%	49%	76%
Clean Acc.	41%	47%	54%	57%	68%	54%	85%

$\epsilon = 2/255$	FCNNA		FCNNB		FCNNC		CNNA		CNNB		CNNC		CNND	
	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2
Bounded MILP	**35%**	**33%**	30%	**37%**	0%	0%	0%	**47%**	0%	49%	0%	0%	0%	0%
AI^2	**35%**	**33%**	**39%**	**37%**	0%	20%	**34%**	**47%**	**19%**	45%	0%	0%	0%	0%
LP-Full	**35%**	**33%**	36%	**37%**	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
DeepPoly	**35%**	**33%**	36%	35%	0%	17%	**34%**	46%	**19%**	45%	0%	0%	0%	0%
Fast-Lin	**35%**	**33%**	35%	34%	0%	30%	33%	45%	14%	43%	**13%**	46%	0%	0%
CROWN	**35%**	**33%**	36%	35%	**1%**	**32%**	**34%**	46%	**19%**	45%	0%	0%	0%	0%
CNN-Cert	**35%**	**33%**	35%	35%	0%	0%	**34%**	46%	**19%**	45%	0%	0%	0%	0%
CROWN-IBP	30%	**33%**	0%	34%	0%	**32%**	0%	46%	0%	46%	0%	50%	0%	47%
IBP	6%	**33%**	0%	35%	0%	**32%**	0%	45%	0%	**49%**	0%	**51%**	0%	**51%**
WK	**35%**	**33%**	35%	34%	0%	30%	33%	45%	14%	43%	**13%**	46%	0%	23%
k-ReLU	**35%**	**33%**	35%	**37%**	0%	0%	0%	44%	0%	0%	0%	0%	0%	0%
RefineZono	0%	**33%**	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
SDPVerify	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
LMIVerify	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Op-norm	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
FastLip	33%	32%	15%	32%	0%	25%	3%	40%	0%	0%	0%	0%	0%	0%
RecurJac	34%	32%	33%	32%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
PGD Upper Bound	36%	35%	41%	39%	26%	38%	43%	49%	52%	50%	49%	53%	62%	54%
Clean Acc.	44%	40%	48%	43%	35%	46%	54%	55%	66%	60%	59%	63%	81%	62%

$\epsilon = 8/255$	FCNNA		FCNNB		FCNNC		CNNA		CNNB		CNNC		CNND	
	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8
Bounded MILP	**19%**	**27%**	1%	**25%**	0%	0%	0%	**34%**	0%	**36%**	0%	0%	0%	0%
AI^2	**19%**	**27%**	**7%**	23%	0%	22%	**8%**	**34%**	0%	20%	0%	14%	0%	0%
LP-Full	15%	**27%**	6%	**25%**	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
DeepPoly	15%	**27%**	6%	20%	0%	22%	**8%**	33%	**1%**	20%	0%	7%	0%	0%
Fast-Lin	15%	25%	4%	18%	0%	19%	3%	26%	0%	15%	0%	7%	0%	0%
CROWN	15%	**27%**	6%	20%	0%	22%	**8%**	33%	**1%**	20%	0%	0%	0%	0%
CNN-Cert	15%	**27%**	5%	20%	0%	0%	7%	33%	0%	20%	0%	0%	0%	0%
CROWN-IBP	9%	**27%**	0%	22%	0%	28%	0%	**34%**	0%	31%	0%	32%	0%	25%
IBP	0%	**27%**	0%	**25%**	0%	**30%**	0%	**34%**	0%	35%	0%	**38%**	0%	**28%**
WK	15%	25%	4%	18%	0%	19%	3%	26%	0%	15%	0%	7%	0%	5%
k-ReLU	15%	**27%**	2%	23%	0%	0%	0%	32%	0%	0%	0%	0%	0%	0%
RefineZono	0%	**27%**	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
SDPVerify	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
LMIVerify	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Op-norm	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
FastLip	12%	**27%**	0%	17%	0%	17%	0%	24%	0%	0%	0%	0%	0%	0%
RecurJac	14%	**27%**	2%	17%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
PGD Upper Bound	22%	28%	23%	26%	19%	34%	34%	34%	33%	39%	36%	40%	41%	31%
Clean Acc.	33%	31%	37%	30%	26%	39%	44%	46%	53%	48%	52%	46%	66%	46%

TABLE IX: *Average running time for single-instance robustness verification* in seconds per correctly-predicted instance on CIFAR-10 models of different verification approaches. The verification is under $\mathcal{L}_\infty$ adversary with ϵ (specified in table) radius. We stop the execution when time exceeds 60 s per instance. We include running time of PGD attack and normal inference as the reference.

	FCNNA	FCNNB	FCNNC	CNNA	CNNB	CNNC	CNND
	clean	clean	clean	clean	clean	clean	clean
Bounded MILP	0.93	10.70	60.00	50.60	60.00	60.00	60.00
AI^2	0.07	2.09	60.00	7.69	57.57	60.00	60.00
LP-Full	0.71	33.68	60.00	60.00	60.00	60.00	60.00
DeepPoly	0.05	0.53	60.00	2.55	41.52	60.00	60.00
Fast-Lin	0.03	0.02	1.03	1.49	11.07	32.62	60.00
CROWN	0.00	0.01	0.08	0.03	0.32	60.00	60.00
CNN-Cert	0.19	0.71	60.00	1.43	35.36	59.69	60.00
CROWN-IBP	0.00	0.00	0.01	0.01	0.01	0.01	0.01
IBP	0.00	0.00	0.01	0.00	0.00	0.01	0.01
WK	0.00	0.01	0.05	0.01	0.06	0.10	54.42
k-ReLU	5.83	20.91	60.00	31.63	60.00	60.00	60.00
RefineZono	60.00	60.00	60.00	60.00	60.00	60.00	60.00
SDPVerify	60.00	60.00	60.00	60.00	60.00	60.00	60.00
LMIVerify	60.00	60.00	60.00	60.00	60.00	60.00	60.00
Op-norm	0.00	0.00	0.00	0.01	0.01	0.03	60.00
FastLip	0.09	0.43	1.35	21.28	49.81	60.00	60.00
RecurJac	1.56	5.67	60.00	57.99	60.00	60.00	60.00
PGD Attack	0.23	0.27	0.35	0.30	0.35	0.39	0.54
Normal Inference	0.00	0.00	0.00	0.00	0.00	0.00	0.00

	FCNNA		FCNNB		FCNNC		CNNA		CNNB		CNNC		CNND	
	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2
Bounded MILP	1.24	1.05	32.64	4.87	60.00	60.00	60.00	4.70	60.00	30.27	60.00	59.55	60.00	60.00
AI^2	1.99	1.06	5.16	7.01	60.00	45.69	23.65	4.81	54.19	27.01	60.00	60.00	60.00	60.00
LP-Full	0.74	0.77	34.64	26.08	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
DeepPoly	0.05	0.04	0.56	0.45	60.00	41.60	2.49	1.10	43.68	17.67	60.00	60.00	60.00	60.00
Fast-Lin	0.01	0.03	0.02	0.05	1.36	1.36	1.48	1.45	11.77	10.92	33.87	34.73	60.00	60.00
CROWN	0.00	0.00	0.01	0.01	0.07	0.04	0.03	0.03	0.33	0.47	60.00	60.00	60.00	60.00
CNN-Cert	0.05	0.17	0.67	0.70	10.71	55.88	1.18	1.26	18.84	33.65	42.77	58.12	60.00	60.00
CROWN-IBP	0.00	0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
IBP	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.01
WK	0.01	0.01	0.01	0.01	0.06	0.03	0.02	0.01	0.08	0.10	0.15	0.09	60.00	0.40
k-ReLU	7.08	7.33	34.73	25.81	60.00	60.00	60.00	26.78	60.00	60.00	60.00	60.00	60.00	60.00
RefineZono	60.00	26.05	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
SDPVerify	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
LMIVerify	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
Op-norm	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.00	0.01	0.01	0.01	0.03	60.00	60.00
FastLip	0.06	0.09	0.30	0.30	1.30	11.06	11.90	11.24	37.44	60.00	60.00	60.00	60.00	60.00
RecurJac	0.90	1.43	4.83	5.14	60.00	60.00	59.51	59.17	60.00	60.00	60.00	60.00	60.00	60.00
PGD Attack	0.24	0.27	0.28	0.30	0.37	0.40	0.30	0.37	0.36	0.45	0.37	0.46	0.55	0.64
Normal Inference	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

	FCNNA		FCNNB		FCNNC		CNNA		CNNB		CNNC		CNND	
	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8
Bounded MILP	3.34	0.95	59.44	3.85	60.00	58.94	60.00	4.55	60.00	27.30	60.00	60.00	60.00	60.00
AI^2	3.68	1.72	47.08	10.74	60.00	36.69	49.39	4.29	60.00	37.84	60.00	57.95	60.00	60.00
LP-Full	1.02	0.82	51.35	20.84	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
DeepPoly	0.05	0.04	0.57	0.29	60.00	25.26	1.95	0.97	38.23	15.16	60.00	57.25	60.00	60.00
Fast-Lin	0.01	0.01	0.03	0.02	1.44	1.58	1.49	1.24	11.77	10.14	38.27	34.86	60.00	60.00
CROWN	0.00	0.00	0.01	0.01	0.07	0.04	0.03	0.03	0.33	0.46	60.00	60.00	60.00	60.00
CNN-Cert	0.04	0.05	0.25	0.50	10.94	42.49	0.48	1.03	4.11	21.88	14.79	41.58	60.00	60.00
CROWN-IBP	0.00	0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
IBP	0.00	0.00	0.00	0.00	0.00	0.01	0.00	0.01	0.01	0.01	0.01	0.01	0.01	0.01
WK	0.01	0.00	0.01	0.01	0.06	0.03	0.02	0.01	0.09	0.11	0.40	0.10	60.00	5.71
k-ReLU	10.14	7.66	57.61	34.26	60.00	60.00	60.00	32.65	60.00	60.00	60.00	60.00	60.00	60.00
RefineZono	60.00	7.77	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
SDPVerify	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
LMIVerify	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00	60.00
Op-norm	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.00	0.01	0.01	0.01	0.03	60.00	60.00
FastLip	0.05	0.06	0.12	0.19	1.44	9.25	8.40	8.36	28.66	59.01	60.00	60.00	60.00	60.00
RecurJac	0.74	0.90	2.81	4.04	48.90	59.14	57.64	58.93	60.00	60.00	60.00	60.00	60.00	60.00
PGD Attack	0.23	0.27	0.27	0.32	0.38	0.39	0.29	0.37	0.38	0.43	0.39	0.47	0.55	0.65
Normal Inference	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

TABLE X: *Average certified robustness radius divided by* 255 on CIFAR-10 models certified by different *deterministic verification approaches*. The verification is under $\mathcal{L}_\infty$ adversary. We include radius of adversarial example found by PGD attack to serve as the upper bound. The **bolded** numbers mark the highest ones among verification approaches.

	FCNN$_A$	FCNN$_B$	FCNN$_C$	CNN$_A$	CNN$_B$	CNN$_C$	CNN$_D$
	clean	clean	clean	clean	clean	clean	clean
Bounded MILP	**5.053**	0.000	0.000	0.000	0.000	0.000	0.000
AI^2	5.047	0.000	0.000	0.000	0.000	0.000	0.000
LP-Full	3.966	0.000	0.000	0.000	0.000	0.000	0.000
DeepPoly	3.984	**2.888**	0.000	**1.669**	0.000	0.000	0.000
Fast-Lin	3.723	2.596	0.812	1.503	0.637	0.000	0.000
CROWN	3.984	**2.888**	**1.065**	**1.669**	**0.765**	0.000	0.000
CNN-Cert	3.893	2.819	0.000	1.625	0.699	0.000	0.000
CROWN-IBP	2.187	0.000	0.000	0.000	0.000	0.000	0.000
IBP	1.026	0.011	0.000	0.000	0.000	0.000	0.000
WK	2.818	1.738	0.000	0.734	0.000	0.000	0.000
k-ReLU	4.203	0.000	0.000	0.000	0.000	0.000	0.000
RefineZono	0.000	0.000	0.000	0.000	0.000	0.000	0.000
SDPVerify	0.000	0.000	0.000	0.000	0.000	0.000	0.000
LMIVerify	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Op-norm	0.183	0.071	0.000	0.002	0.001	0.000	0.000
FastLip	2.988	1.256	0.000	0.599	0.000	0.000	0.000
RecurJac	3.596	2.400	0.000	0.000	0.000	0.000	0.000
PGD Upper Bound	5.679	5.521	6.037	3.482	2.560	4.187	1.819

	FCNN$_A$		FCNN$_B$		FCNN$_C$		CNN$_A$		CNN$_B$		CNN$_C$		CNN$_D$	
	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2
Bounded MILP	**6.429**	**9.301**	0.000	5.229	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
AI^2	6.033	**9.301**	0.000	0.185	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
LP-Full	4.930	8.747	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
DeepPoly	5.009	7.408	**3.683**	4.720	0.000	0.173	**2.601**	4.682	0.000	0.498	0.000	0.000	0.000	0.000
Fast-Lin	4.714	7.103	3.211	4.036	0.655	3.080	2.269	4.079	1.294	2.486	0.000	0.000	0.000	0.000
CROWN	5.009	7.408	**3.683**	4.720	**0.847**	**4.071**	**2.601**	4.682	**1.551**	3.030	0.000	0.000	0.000	0.000
CNN-Cert	4.856	7.321	3.575	4.911	0.000	0.000	2.532	4.578	1.339	2.702	0.000	0.000	0.000	0.000
CROWN-IBP	2.717	7.863	0.000	5.004	0.000	3.936	0.000	5.415	0.000	3.889	0.000	3.909	0.000	3.551
IBP	0.826	8.324	0.016	**5.027**	0.000	3.508	0.028	**5.958**	0.000	**4.611**	0.000	**4.411**	0.000	**4.233**
WK	3.939	7.103	2.407	4.036	0.000	3.080	1.476	4.079	0.423	2.486	0.000	0.000	0.000	0.000
k-ReLU	5.184	8.871	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
RefineZono	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
SDPVerify	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
LMIVerify	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Op-norm	0.174	0.011	0.068	0.000	0.000	0.000	0.002	0.000	0.001	0.000	0.000	0.000	0.000	0.000
FastLip	3.696	6.319	1.359	3.399	0.000	2.404	0.742	3.523	0.000	0.000	0.000	0.000	0.000	0.000
RecurJac	4.550	6.375	2.765	3.509	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
PGD Upper Bound	7.856	13.485	8.581	10.864	9.292	9.181	7.185	11.432	6.629	10.301	7.901	9.578	5.472	9.981

	FCNN$_A$		FCNN$_B$		FCNN$_C$		CNN$_A$		CNN$_B$		CNN$_C$		CNN$_D$	
	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8
Bounded MILP	**12.134**	**22.428**	0.000	**23.259**	0.000	0.000	0.000	18.953	0.000	1.328	0.000	0.000	0.000	0.000
AI^2	10.972	22.364	0.000	13.613	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
LP-Full	8.542	21.593	0.000	10.625	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
DeepPoly	8.882	20.870	**5.398**	15.780	0.000	6.411	**5.156**	11.617	0.000	7.294	0.000	0.000	0.000	0.000
Fast-Lin	7.825	17.946	4.637	11.422	0.881	9.259	4.471	9.171	2.396	6.469	0.000	0.000	0.000	0.000
CROWN	8.882	20.870	**5.398**	15.780	**1.226**	11.615	**5.156**	11.617	**2.890**	8.093	0.000	0.000	0.000	0.000
CNN-Cert	8.444	20.163	5.169	14.900	0.000	2.452	4.952	11.601	2.523	7.019	0.000	0.000	0.000	0.000
CROWN-IBP	5.614	21.014	0.108	20.603	0.000	14.852	0.589	15.012	0.000	14.604	0.000	13.934	0.000	11.710
IBP	1.517	22.276	0.074	22.271	0.000	**15.880**	0.277	**19.115**	0.000	**18.770**	0.000	**18.920**	0.000	**15.553**
WK	7.003	17.946	3.877	11.422	0.000	9.259	3.577	9.171	1.541	6.469	0.000	5.137	0.000	0.000
k-ReLU	9.304	22.300	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
RefineZono	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
SDPVerify	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
LMIVerify	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Op-norm	0.117	0.020	0.052	0.000	0.000	0.000	0.002	0.000	0.001	0.000	0.000	0.000	0.000	0.000
FastLip	6.309	18.002	1.992	10.019	0.000	8.799	2.173	8.147	0.000	0.000	0.000	0.000	0.000	0.000
RecurJac	7.539	18.002	3.971	10.027	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
PGD Upper Bound	15.500	35.723	15.783	35.278	19.558	25.796	15.219	31.399	13.128	34.687	14.913	37.088	12.040	28.578

TABLE XI: *Average running time for robustness radius computation* in seconds per correctly-predicted instance on CIFAR-10 models of different verification approaches. The verification is on $\mathcal{L}_\infty$ ball. We stop the execution when time exceeds 120 s per instance. We include running time of PGD attack as the reference.

	FCNNA	FCNNB	FCNNC	CNNA	CNNB	CNNC	CNND
	clean	clean	clean	clean	clean	clean	clean
Bounded MILP	23.76	120.00	120.00	120.00	120.00	120.00	120.00
AI^2	40.75	120.00	120.00	120.00	120.00	120.00	120.00
LP-Full	8.96	120.00	120.00	120.00	120.00	120.00	120.00
DeepPoly	0.40	5.26	120.00	24.44	120.00	120.00	120.00
Fast-Lin	0.05	0.22	12.19	13.14	112.11	120.00	120.00
CROWN	0.03	0.05	0.43	0.20	2.96	120.00	120.00
CNN-Cert	0.24	3.05	120.00	5.03	102.79	120.00	120.00
CROWN-IBP	0.02	0.02	0.04	0.03	0.04	0.05	0.06
IBP	0.02	0.02	0.04	0.02	0.04	0.05	0.08
WK	0.03	0.05	0.25	0.11	0.70	120.00	120.00
k-ReLU	100.18	120.00	120.00	120.00	120.00	120.00	120.00
RefineZono	120.00	120.00	120.00	120.00	120.00	120.00	120.00
SDPVerify	120.00	120.00	120.00	120.00	120.00	120.00	120.00
LMIVerify	120.00	120.00	120.00	120.00	120.00	120.00	120.00
Op-norm	0.00	0.00	0.00	0.01	0.01	0.03	120.00
FastLip	0.41	1.98	13.48	77.41	120.00	120.00	120.00
RecurJac	5.72	26.99	120.00	120.00	120.00	120.00	120.00
PGD Attack	2.54	2.87	3.68	2.89	3.48	3.69	4.81

	FCNNA		FCNNB		FCNNC		CNNA		CNNB		CNNC		CNND	
	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2	adv2	cadv2
Bounded MILP	26.48	16.01	120.00	117.08	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
AI^2	59.12	48.56	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
LP-Full	8.89	8.65	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
DeepPoly	0.41	0.36	5.07	3.91	120.00	120.00	22.75	11.02	120.00	120.00	120.00	120.00	120.00	120.00
Fast-Lin	0.05	0.08	0.22	0.27	12.10	14.38	13.27	13.08	108.53	96.77	120.00	120.00	120.00	120.00
CROWN	0.03	0.03	0.05	0.05	0.43	0.41	0.20	0.20	3.00	4.26	120.00	120.00	120.00	120.00
CNN-Cert	0.25	0.34	3.17	2.66	120.00	120.00	5.60	5.75	113.73	116.81	120.00	120.00	120.00	120.00
CROWN-IBP	0.02	0.03	0.02	0.03	0.05	0.06	0.03	0.04	0.04	0.10	0.05	0.07	0.06	0.08
IBP	0.02	0.02	0.02	0.03	0.04	0.05	0.03	0.04	0.04	0.05	0.04	0.05	0.08	0.10
WK	0.03	0.04	0.05	0.07	0.25	0.28	0.11	0.13	0.79	1.35	120.00	120.00	120.00	120.00
k-ReLU	91.47	93.79	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
RefineZono	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
SDPVerify	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
LMIVerify	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
Op-norm	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.03	0.03	120.00	120.00
FastLip	0.42	0.40	1.88	1.74	16.78	65.67	99.47	65.69	120.00	120.00	120.00	120.00	120.00	120.00
RecurJac	5.83	6.41	27.72	28.83	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
PGD Attack	2.79	3.64	3.04	3.60	4.15	4.26	3.19	4.13	3.79	4.57	4.01	4.73	5.05	6.63

	FCNNA		FCNNB		FCNNC		CNNA		CNNB		CNNC		CNND	
	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8	adv8	cadv8
Bounded MILP	36.26	19.76	120.00	89.66	120.00	120.00	120.00	106.38	120.00	120.00	120.00	120.00	120.00	120.00
AI^2	89.11	68.36	120.00	117.86	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
LP-Full	9.56	8.43	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
DeepPoly	0.47	0.38	5.10	2.87	120.00	120.00	18.33	9.65	120.00	119.79	120.00	120.00	120.00	120.00
Fast-Lin	0.05	0.06	0.23	0.23	12.23	12.47	12.97	13.05	106.36	92.09	120.00	120.00	120.00	120.00
CROWN	0.03	0.03	0.05	0.05	0.43	0.41	0.20	0.20	3.00	4.15	120.00	120.00	120.00	120.00
CNN-Cert	0.28	0.27	3.56	3.52	120.00	120.00	6.38	6.27	112.92	118.42	120.00	120.00	120.00	120.00
CROWN-IBP	0.02	0.03	0.02	0.03	0.05	0.06	0.03	0.04	0.04	0.09	0.05	0.06	0.06	0.08
IBP	0.02	0.03	0.03	0.03	0.04	0.05	0.03	0.03	0.04	0.05	0.05	0.06	0.08	0.09
WK	0.03	0.04	0.05	0.07	0.24	0.27	0.10	0.13	0.74	1.19	120.00	1.61	120.00	120.00
k-ReLU	96.06	101.99	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
RefineZono	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
SDPVerify	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
LMIVerify	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
Op-norm	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.00	0.01	0.01	0.03	0.03	120.00	120.00
FastLip	0.45	0.45	2.05	1.79	17.55	86.96	102.45	70.16	120.00	120.00	120.00	120.00	120.00	120.00
RecurJac	6.91	7.11	30.71	34.06	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00	120.00
PGD Attack	3.41	4.58	3.80	5.17	4.95	5.52	4.04	5.66	4.48	6.73	4.58	6.66	5.67	8.05

the results of Bounded MILP, AI^2, CROWN, and CNN-Cert). However, if measured by average certified robustness radius, we can observe that complete verification approaches can certify much larger radius since they do not use any relaxations (see table X, the result that Bounded MILP $\approx$ AI^2 $>$ CROWN $\approx$ CNN-Cert).

From the running time statistics, we further observe that a main cause of the failing cases (i.e., 0 robust accuracy or certified radius) for these approaches is the excessive verification time. In particular, since the evaluation of average certified robustness radius requires multiple times of verification for each instance, it suffers more severely from long verification time.

References

[1] A. Boopathy, T.-W. Weng, P.-Y. Chen, S. Liu, and L. Daniel, "Cnn-cert: An efficient framework for certifying robustness of convolutional neural networks," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, 2019, pp. 3240–3247.

[2] (2020) Welcome to cvxpy 1.1 - cvxpy 1.1 documentation. https://www.cvxpy.org/.

[3] M. Fazlyab, M. Morari, and G. J. Pappas, "Safety verification and robustness analysis of neural networks via quadratic constraints and semidefinite programming," *arXiv preprint arXiv:1903.01287*, 2019.

[4] T. Gehr, M. Mirman, D. Drachsler-Cohen, P. Tsankov, S. Chaudhuri, and M. Vechev, "AI^2: Safety and robustness certification of neural networks with abstract interpretation," in *2018 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2018, pp. 3–18.

[5] S. Gowal, K. D. Dvijotham, R. Stanforth, R. Bunel, C. Qin, J. Uesato, R. Arandjelovic, T. Mann, and P. Kohli, "Scalable verified training for provably robust image classification," in *Proceedings of the IEEE International Conference on Computer Vision*, 2019, pp. 4842–4851.

[6] (2020) Gurobi - the fastest solver - gurobi. https://www.gurobi.com/.

[7] (2020) Keras: the python deep learning api. https://keras.io/.

[8] A. Krizhevsky, G. Hinton *et al.*, *Learning multiple layers of features from tiny images*. Citeseer, 2009.

[9] M. Lecuyer, V. Atlidakis, R. Geambasu, D. Hsu, and S. Jana, "Certified robustness to adversarial examples with differential privacy," in *2019 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2019, pp. 656–672.

[10] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *International Conference on Learning Representations*, 2018. [Online]. Available: https://openreview.net/forum?id=rJzIBfZAb

[11] (2020) Onnx home. https://onnx.ai/.

[12] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga *et al.*, "Pytorch: An imperative style, high-performance deep learning library," in *Advances in neural information processing systems*, 2019, pp. 8026–8037.

[13] A. Raghunathan, J. Steinhardt, and P. S. Liang, "Semidefinite relaxations for certifying robustness to adversarial examples," in *Advances in Neural Information Processing Systems*, 2018, pp. 10 877–10 887.

[14] H. Salman, G. Yang, H. Zhang, C.-J. Hsieh, and P. Zhang, "A convex relaxation barrier to tight robustness verification of neural networks," in *Advances in Neural Information Processing Systems*, 2019, pp. 9832–9842.

[15] G. Singh, R. Ganvir, M. Püschel, and M. Vechev, "Beyond the single neuron convex barrier for neural network certification," in *Advances in Neural Information Processing Systems*, 2019, pp. 15 072–15 083.

[16] G. Singh, T. Gehr, M. Püschel, and M. Vechev, "An abstract domain for certifying neural networks," *Proceedings of the ACM on Programming Languages*, vol. 3, no. POPL, p. 41, 2019.

[17] G. Singh, T. Gehr, M. Püschel, and M. Vechev, "Robustness certification with refinement," in *International Conference on Learning Representations*, 2019. [Online]. Available: https://openreview.net/forum?id=HJgeEh09KQ

[18] C. Szegedy, W. Zaremba, I. Sutskever, J. Bruna, D. Erhan, I. Goodfellow, and R. Fergus, "Intriguing properties of neural networks," *arXiv preprint arXiv:1312.6199*, 2013.

[19] (2020) Tensorflow. https://www.tensorflow.org/.

[20] V. Tjeng, K. Y. Xiao, and R. Tedrake, "Evaluating robustness of neural networks with mixed integer programming," in *International Conference on Learning Representations*, 2019. [Online]. Available: https://openreview.net/forum?id=HyGIdiRqtm

[21] L. Weng, H. Zhang, H. Chen, Z. Song, C.-J. Hsieh, L. Daniel, D. Boning, and I. Dhillon, "Towards fast computation of certified robustness for relu networks," in *International Conference on Machine Learning*, 2018, pp. 5276–5285.

[22] E. Wong and Z. Kolter, "Provable defenses against adversarial examples via the convex outer adversarial polytope," in *International Conference on Machine Learning*, 2018, pp. 5286–5295.

[23] E. Wong, F. Schmidt, J. H. Metzen, and J. Z. Kolter, "Scaling provable adversarial defenses," in *Advances in Neural Information Processing Systems*, 2018, pp. 8400–8409.

[24] H. Zhang, T.-W. Weng, P.-Y. Chen, C.-J. Hsieh, and L. Daniel, "Efficient neural network robustness certification with general activation functions," in *Advances in neural information processing systems*, 2018, pp. 4939–4948.

[25] H. Zhang, P. Zhang, and C.-J. Hsieh, "Recurjac: An efficient recursive algorithm for bounding jacobian matrix of neural networks and its applications," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 33, 2019, pp. 5757–5764.

[26] H. Zhang, H. Chen, C. Xiao, S. Gowal, R. Stanforth, B. Li, D. Boning, and C.-J. Hsieh, "Towards stable and efficient training of verifiably robust neural networks," in *International Conference on Learning Representations*, 2020. [Online]. Available: https://openreview.net/forum?id=Skxuk1rFwB